Predicting Flight Delays the Night Before

Leakage-free feature engineering, calibrated classifiers, and what-if rescheduling across 336k departures.

CORE FOCUS

Strict Day-Ahead Operational Decision Boundary

DATASET

EWR • JFK • LGA

MODELS

Logistic Regression • XGBoost

01 / OPERATIONAL CHALLENGE

The Real-World Decision Boundary

The Core Question

Can we reliably flag flights landing 15+ minutes late before operations begin tomorrow?

The Common Pitfall

Most academic benchmarks use departure delays (corr = 0.915) or live gate data. This leaks the future.

Our Strict Rule

Use only published schedule metrics and prior-day history available by 23:59.



02 / EXPLORATORY DATA ANALYSIS

Delay Dynamics Across 336,776 Flights

10% → 37%

Cascade Effect

Evening flights absorb cumulative aircraft rotation delays.

0.40

Lag-1 Autocorrelation

Disruption carries over to tomorrow, then drops sharply to 0.13.

18% vs 34%

Severe Seasonality

Fall months are peaceful; December winter peaks in disruption.

Note: 9,430 cancelled or diverted flights excluded from target training, but preserved for advance schedule congestion counts.

03 / FEATURE ENGINEERING

35 Engineered Clues Built Past-Only

Calendar & Cyclic

Continuous sin/cos time encodings, day of week, and holiday distance.

Traffic & Congestion

Scheduled departures per origin-day, hourly volume, and rotation leg.

Smoothed History

Route, carrier, and aircraft past rates shrunk with Beta(m=20) prior.

Automated Test

Verified: scrambling future outcomes does not alter past features.

Design Rule: Month and Day-of-Year features were strictly removed to prevent tree models from memorizing specific calendar days.

04 / EXPERIMENTAL DESIGN

Strict Chronological Evaluation

PHASE 1 • TRAINING

January – September

244,737 flights • Base delay rate: 25.0%

PHASE 2 • VALIDATION

Oct – Nov

55,589 flights • Rate: 17.8%

PHASE 3 • TEST

December

27,020 flights • Rate: 33.5%

Crucial Regularization Finding in XGBoost

With default leaves, XGBoost memorized days via constant daily features. Setting min_child_weight = 2000 forced leaves to cover ≥8,000 flights (~9 days of NYC traffic), stopping memorization.

Zero data peeking: all model parameters, thresholds, and calibrators were selected strictly on validation months.

05 / MODEL DISCRIMINATION

Significant Uplift Above Baseline



Both classifiers outperform route history by +22% in PR-AUC on held-out December.

Unobservable day-level weather sets the upper ceiling on ROC-AUC (~0.65).

06 / OPERATIONAL DECISIONS

Enforcing High Precision Thresholds

66.1%

Validated Precision Target

When XGBoost flags a flight as high risk, 2 out of 3 flights are genuinely delayed 15+ minutes.

The Precision Constraint

Airlines cannot afford high false alarms. Operating points required ≥50% precision on validation.

The Recall Trade-Off

Under fixed calibration, recall was only 3.5% (flags 1.8% of flights) because December was severe.

Actionable Scope

Enables high-conviction alerts for dispatchers rather than broad uncertain warnings.

Operating threshold fixed at 0.498 from validation metrics; applied cold to December test set.

07 / RELIABILITY & LABEL SHIFT

Solving the Seasonal Calibration Drift

THE BREAKDOWN

Fixed Platt Scaling

Calibrators fitted on calm Oct-Nov (17.8% delays) systematically under-predicted stormy December (33.5%).

ECE: 0.102

Mean predicted risk dropped to 23.4%

THE SOLUTION

Rolling 28-Day Recalibration

Refitting daily on previous 4-week realized outcomes tracks environmental drift dynamically.

ECE: 0.047

Halved calibration error; recall rose to 6.2%

Practical takeaway: Risk models in seasonal transportation require rolling posterior recalibration.

08 / PRESCRIPTIVE SIMULATION

Can We Reschedule High-Risk Flights?

Risk Threshold	Flagged Flights	Observed Late	Moved (±2h)	Risk Before → After
p ≥ 0.40	5,750	47.4%	295	0.504 → 0.444
p ≥ 0.50 (Primary)	**1,979**	**52.0%**	**133 (98 moved earlier)**	**0.571 → 0.510**
p ≥ 0.60	473	55.6%	33	0.657 → 0.595

Simulation Takeaway: Moving flights earlier (-2h) yields small risk reductions. However, systemic delay risk is anchored primarily to day-level conditions and route congestion rather than the departure hour alone.

09 / CRITICAL ASSUMPTIONS

Model Boundaries & Constraints



1. Weather Data Cap

Day-level storm systems explain core variance. Without day-ahead forecast ingestion, ROC-AUC caps at ~0.65.

2. Single-Year Corridor Scope

Only NYC departures in 2013: inbound rotation legs into NYC remain unobserved.

3. Associational Rescheduling

What-if re-timing ignores crew duty time limits, gate constraints, and passenger connection banks.

Acknowledging constraints strengthens experimental credibility during presentation defense.

Phase 1 Key Takeaways

Zero-Leakage Soundness

Strict temporal split and automated leakage tests guarantee deployable, honest predictions.

Model Equivalence

LR (0.461) and XGBoost (0.468) perform similarly once boosting tree leaves are properly regularized.

Adaptive Recalibration

A rolling 28-day window overcomes seasonal base-rate shifts, cutting probability error in half.

Phase 2 Direction

Next step: Compact Multi-Layer Perceptron (MLP) learning entity embeddings for carriers and airports.

All code, models, and reproducible notebooks are cataloged on GitHub.



Thank You

Day-ahead flight delay risk prediction with leakage-free features and rolling probability calibration.

Open for Questions & Discussion

Repository: github.com/techatanu/Flight_Delay_Risk